

AI Is Here To Stay: Misinformation and Human-Centric Models Between Risks and Opportunities

MARCO BELLÒ, University of Padua, Italy

For more than half a century, artificial intelligence remained a matter of speculation for novelists and philosophers. This changed in 2017 due to the introduction of the transformer architecture, spearheading the AI boom whose effects are still scarcely understood. This paper explores potential risks and use cases of this technology, with a "human-centric" cut: the first half showcases generation and detection of fake news and deepfakes, as well as how much LLMs can influence people's opinions and beliefs. The focus then shifts to the AIs themselves, about their own biases and their susceptibility to external stimuli, concluding with a quick overview on the state of research about AIs for positive social impact.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Artificial intelligence**; **Natural language processing**; **Reasoning about belief and knowledge**; Cognitive science.

Additional Key Words and Phrases: AI, NLP, Misinformation, Deepfakes, Opinions, Biases, AI4SI

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1 INTRODUCTION

Politicians are some of the most influential people in human society: from the greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to steer society. Shapiro and Page in 1984 showed how nearly all 20th century U.S. Presidents have tried to lead public opinion, with varying degrees of success. [6]. As societies become more democratic, the leaders' responsibility to convince potential followers that their policies are beneficial become greater, making speeches play a vital role in the functioning of a modern society [1]: leaders depend on the verbal power to persuade people [2].

Moreover, democratic stability depends on citizens on the losing side accepting election outcomes. Clayton et. al. evaluated the effects of exposure to multiple statements from (at the time) former president Donald Trump attacking the legitimacy of the 2020 US presidential election. They found no evidence indicate that elites can erode democratic norms easily or that the effects of norm violations are uniform across the entire population. However, elite rhetoric can shape normative beliefs in core democratic values such as confidence in elections and support for peaceful transfers of power. This was especially evident amongst people who approved of Trump's performance in office [3].

Broadly speaking, we can distinguish between two types of speeches: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social networks posts. This last category is especially interesting since it allows politicians a direct and

Author's address: Marco Bellò, marco.bello.3@studenti.unipd.it, University of Padua, Padua, Italy.

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high-volume communication channel to their potential followers: U.S. President Donald Trump tweeted 57.0000 times between May 2009 and January 2021 [?] alone, while Italy's Prime Minister Giorgia Meloni totalled more than eighty million social networks interactions in 2025 [?].

On the other hand, internal speeches, while being easily accessible in public state-managed databases, pose some real challenges: they are more technical and codified, and until 1985 Italian stenographic documentation did not report speeches word for word, but rather translated into a "language of the parliament" [4, 5].

2 AI FOR FAKE NEWS GENERATION AND DETECTION

Fake news has rapidly become a significant concern in the digital age, thanks to their virality and potential damages. They spread faster and generate more engagement than truthful information [?], and can influence public opinion, manipulate elections and pose a threat to public health. For example, the World Economic Forum has identified the proliferation of false content as the leading short-term global risk in 2025 [?], and a BBC investigation found Russian-funded fake news networks aiming to disrupt European elections [?]. Moreover, fake news on health can cause psychological disorders such as panic, fear, depression, and fatigue [?], making the World Health Organization to call for development of international fact-checking organizations to combat this phenomenon [?].

Adding to the problem, the recent advancements in generative artificial intelligence have made it significantly easier to propagate misinformation through the web: generated content is increasingly indistinguishable from human-written text, sometimes even perceived as more credible [?], citing true evidence to support false claims [?], and inducing the illusion of majority opinion thanks to the sheer volume of information produced [?].

That being said, not all findings are entirely negative: Drolsbach and Pröllochs [?] shift their focus from potential societal consequences to real-world prevalence, conducting a large-scale analysis on the platform X. They analyzed a dataset comprising 91.452 misleading posts, both human and AI-generated, flagged through X's *Community Notes* platform¹. Their findings reveal that generated fake news are often centered on entertaining content rather than controversial or political subjects, and tends to exhibit a more positive sentiment than conventional forms of misinformation. Unfortunately, it is also significantly more likely to go viral.

Lastly, AI agents can produce more than just text: they can create realistic images, videos and sounds, allowing them to make digital copies of real or fictional people, known as *deepfakes*. In March 2019, this technology has been used to trick a UK-based energy firm's CEO into transferring \$243.000 to a malicious party, disguised as an entirely AI-generated executive from their parent company [?]. Deepfakes also increase the amount of conspiratorial videos on the internet, and they are especially vicious when targeting children, whose worldviews are easily swayed by deceptive, highly photorealistic content [?].

It follows that detecting and mitigating fake news is crucial. From the foundational work by Devlin et al. on *BERT* in 2018 [?], which revolutionized natural language processing through deep bidirectional transformers, to the application of said transformers in identifying automatically generated headlines [?]; the landscape of automated fake news detection has significantly expanded. Vijjali et al. [?] developed a two-stage transformer-based model for detecting COVID-19 related misinformation, combining fact-checking with textual entailment² to verify claims. Their model performs significantly better than other baseline NLP approaches (table 1).

¹Community Notes, formerly known as Birdwatch, is community-driven content moderation program on X (formerly Twitter), where contributors can add context such as fact-checks under a post, image or video. GitHub repository: <https://github.com/twitter/communitynotes>

²In natural language processing, textual entailment is a directional relation between text fragments. If the first sentence is true and that makes the second sentence true as well, then the first sentence entails the second.

Table 1. Precision metrics for two-stage transformer model for COVID-19 fake news detection. Vijjali et al [?]]

Models	MRR	Recall@10	Accuracy
TF-IDF	0.477	0.635	0.525
GloVe	0.182	0.410	0.579
MobileBERT	0.561	0.735	0.710
BERT	0.632	0.795	0.810
ALBERT	0.582	0.675	0.825
BERT+ALBERT	0.632	0.795	0.855

Jwa et al. [?] propose an improved *exBAKE* model that leverages pre-training on a BERT model to accurately understand and assess articles' authenticity. They only analyzed the relationship between headlines and body text. Results are shown in table 2.

Table 2. Precision metrics for exBAKE transformer model on fake news recognition. Jwa et al. [?]]

Models	F1	AGR	DSG	DSC	UNR
Majority vote	0.210	0.000	0.000	0.000	0.839
BERT	0.656	0.651	0.145	0.839	0.989
BAKE	0.734	0.667	0.463	0.822	0.986
exBAKE	0.746	0.684	0.501	0.813	0.988
Upper bound	0.754	0.588	0.667	0.765	0.997

Schütz et al. [?] experimented on *FakeNewsNet* dataset with *XLNet*, BERT, *RoBERTa*, *DistilBERT*, and *ALBERT* and various combinations of hyperparameters. The evaluation shows that titles are enough to attain 85% accuracy, while concatenating them with the body text increases it to 87%. Lastly, on the matter of deepfakes, Bansal et al. [?] use *Convolutional Neural Networks* (CNN) and *Deep Convolutional Generative Adversarial Networks* (GAN) to detect them with high accuracy, as shown in figure 1.

These were just a small selection of the many research works in the field of AI-aided fake news generation and detection, which while being extremely relevant and proliferous, are but a fraction of the many potential uses for these technologies.

3 AI FOR UNDERSTANDING AND INFLUENCING HUMAN BEHAVIOUR

Today's society is already fully dependent on technology: from the banking and traffic infrastructures to public healthcare, IT systems have become essential. Individuals are in the same situation: virtually everyone in the Global North³ under the age of 65 possess a smartphone and use it daily [?]. It follows that artificial intelligence will become an integral part of our personal and professional lives, therefore modeling it to mimic our behaviour could increase its usefulness and understandability. There is evidence that humans can exploit it to acquire better comprehension of a phenomenon [?], and that it can enhance creativity in heterogeneous groups [?]. Moreover, LLMs represent a significant methodological shift in computational communication science, enabling a more flexible, more nuanced, but

³The Global North is a collection of countries corresponding to the northern hemisphere which the UNCTAD describes as broadly comprising Northern America and Europe, Israel, Japan, South Korea, Australia, and New Zealand [?]

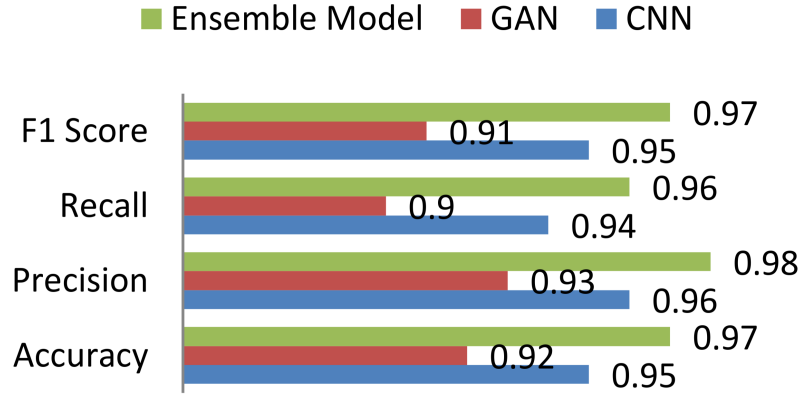


Fig. 1. Transformer models scores on deepfakes detection. Bansal et al. [?]

also less controllable exploration of social theories that have historically been difficult to reduce to simple mathematical formalisms [?]. Overall, AI promises to be a good fit for understanding, modeling, and replicating human behaviour.

For these very purposes, Dos Santos Melicio et al. [?] propose a three-phase AI framework (figure 2) that analyzes verbal and non-verbal social cues. First, deep learning methods are employed to extract relevant information from the environment. Then, the episode detection phase uses extracted features to identify key moments, or episodes, which are then classified in the activity interpretation phase. Overall, the system can recognize verbal hints with 87% accuracy and non-verbal ones with 89% accuracy, with zero false positives. The system was tested in an automated assessment of communicative skills of children with Autism Spectrum Disorder (ASD), a challenging context where social cues may be less noticeable or outright absent.



Fig. 2. Composite AI framework: Deep Learning methods extract features which are then combined with rule-based systems for detecting key episodes, and then classifiers are used to interpret activities happening in the scene. Dos Santos Melicio et al. [?]

Another possible use of such capabilities is hate speech detection: the rise of social networks and online platforms translated into a surge in hate speech across geographical and cultural boundaries. In recent studies, approximately 30% of the adolescents surveyed reported experiencing cyberbullying at some point in their lives. Furthermore, around 13% indicated that they had been cyberbullied within the 30 days before the survey [?].

To address these challenges, Chapagain et al. [?] evaluate different LLMs (BART, ELECTRA, BERT, RoBERTa, and GPT-2) on the extensive *MetaHate* dataset [?]. ELECTRA achieved the highest F1 score (table 3), outperforming all other baselines in hate speech classification.

AI technologies can also be used to influence people's opinions. Huq et al. [?] proved it by evaluating AI-assisted messaging in an online chat platform. 557 Participants were randomly assigned to sessions of six to fifteen people, further subdivided into groups of two to four. They discussed politically controversial topics within three-minutes

Table 3. Performance of classifiers on MetaHate. Chapagain et al [?] .

Models	F1 Score	Accuracy
SVM	0.8380	0.8466
CNN	0.8422	0.8612
BERT	0.8809	0.8879
GPT2	0.6504	0.6152
T5	0.8707	0.8625
DeBERTa	0.8808	0.8746
Longformer	0.8845	0.8785
RoBERTa	0.8908	0.8858
XLNet	0.8917	0.8870
BART	0.8928	0.8886
DistilBERT	0.8940	0.8905
ELECTRA	0.8980	0.8946

sessions, at the end of each they chose whether to remain in their current group, join another, or create a new one. Some of them received suggestions from large language models, either personalized to their own opinion ("*individuals*") or more similar to the group's ("*relational*").

The results show that individual assistance amplified communication volume yet increased separation between groups, while relational assistance fostered more receptive conversations and produced more heterogeneous, cross-cutting group configurations, highlighting both the dangers and the possibilities of employing artificial agents in such a fashion.

Similar conclusions can be seen in other research results: Hohenstein et al. [?] highlights how AI response suggestion systems change how people interact with and perceive one another in both pro-social and anti-social ways. Moreover, Noy et al. [?] show that people tend to send more messages when suggestions are available, but rarely edit them, suggesting partial delegation of expressive effort.

Overall, AI tools seem capable of reducing harmful online behaviors, but can potentially be extremely disruptive due to their capability of influencing people's opinions and reducing autonomous behaviour.

4 AI OWN BIASES AND INFLUENCEABILITY

Since artificial agents are trained on mostly human-generated data, they learn human biases and tendencies themselves, developing both historical and political preferences in the textual content they generate [? ? ?]. They tend to exhibit social sycophancy, meaning they agree with and flatter the user at the cost of correctness [?], and they are easily influenced by small changes in prompt wording [? ?]. This can raise ethical concerns as biased AIs lead to discrimination or exclusion of marginalized groups [?].

Rozado, in the context of US politics, measured the political bias of many popular large language models [?]. First, they calculated the similarity between AI-generated text and public speeches from Congress representatives (both Democrat and Republican). Then, they used an LLM to annotate as left- or right-leaning AI-generated policy recommendations. Consequently, they did a sentiment analysis on AI-generated comments about american public figures, such as legislators or journalists (figure 3). Lastly, they administered three different political orientation tests to the various LLMs. The results show substantial evidence that the models are biased and left-leaning. Table 4 shows the three most and three least biased LLMs among those tested.

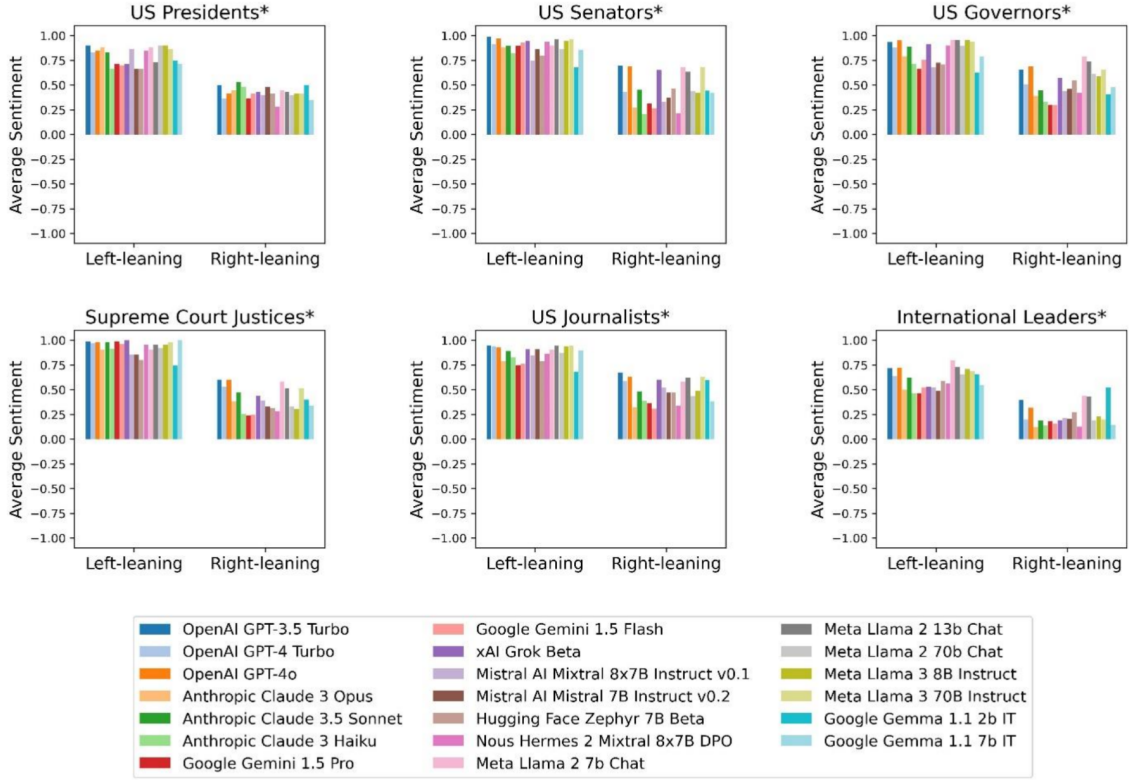


Fig. 3. Average sentiment (negative: -1, neutral: 0, positive: 1) towards ideologically aligned public figures in conversational LLMs' generated texts. Statistically significant two-sample t-tests at the 0.01 threshold are indicated with an asterisk. Rozado [?]

Table 4. Ranking of political bias in conversational LLMs sorted in ascending order from least politically biased to most. Rozado [?]

Rank	Model
1	Google Gemma 1.1 2b IT
2	xAI Grok Beta
3	Mistral AI Mistral 7B Instruct v0.2
...	...
18	Nous Hermes 2 Mixtral 8x7B DPO
19	Google Gemini 1.5 Pro
20	Google Gemini 1.5 Flash

As previously stated, AIs can be easily influenceable. Mehdizadeh and Hilbert [?] test such a theory by subjecting LLMs to peer-pressure. The models are immersed into fictitious social networks (figure 4), and each agent periodically receives a natural language prompt summarizing the opinions of its immediate neighbors. It then decides whether to update its opinions or not. The flowchart of the simulation can be seen at figure 5.

Overall, the study highlights two main results: agents' malleability depends on the model (*Gemini 1.5 Flash* requires over 70% peer disagreement to flip, whereas *ChatGPT-4o-mini* shifts with a dissenting minority), and different cognitive

orientations respond differently to outside stimuli. For *values* and *opinions*, agents show a strong resistance to abandoning a "Yes" stance, making them robust once affirmed, whereas for *attitudes* and *intentions* the greatest challenge is overcoming a negative "No". On the other hand, *beliefs* display a near-perfect symmetry.

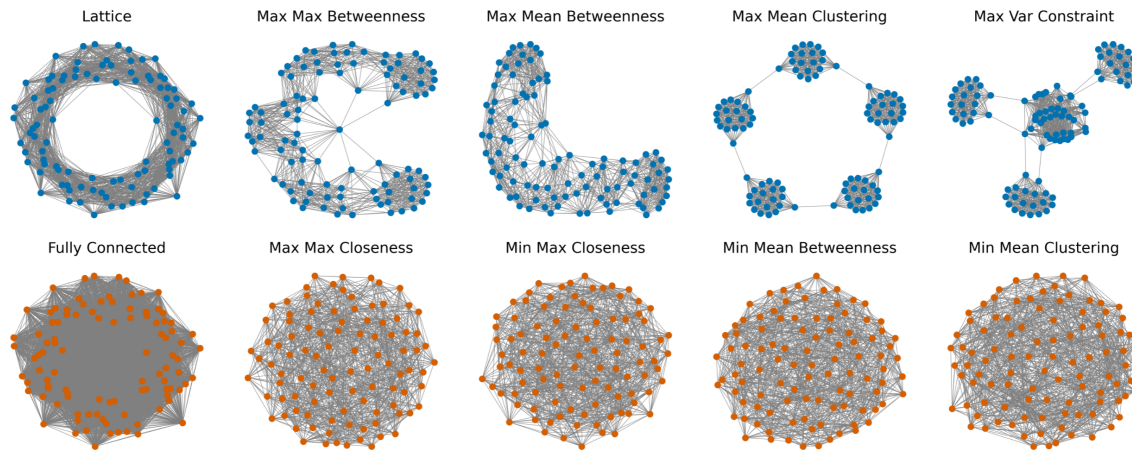


Fig. 4. The ten 100-nodes networks used in the study. Mehdizadeh and Hilbert [?].

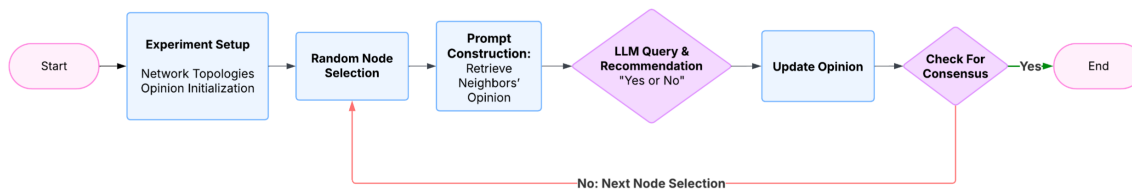


Fig. 5. Flowchart of the simulation procedure in the LLM-driven network model. The process continues asynchronously across the network until a specified number of steps or convergence criterion is met. The setup measures how local peer influence, mediated through LLM interpretation, shapes the emergent collective opinion. Mehdizadeh and Hilbert [?]

In conclusion, the literature offers empirical proof that artificial agents inherit biases from their training data and are influenced by external stimuli. This, combined with their ability to manipulate humans' opinions (section 3), raises ethical concerns that must be addressed.

5 ETHICAL AI FOR POSITIVE SOCIAL IMPACT

Up until this point, this paper highlighted how powerful and dangerous AI technologies can be. Unfortunately, prohibiting their development is not an option: all major global players are heavily investing in this technology. For example, its State Council aims China to be the global leader in the development of artificial intelligence theory and technology by 2030 [?], and Microsoft has announced an investment of thirty billion dollars in the UK on AI infrastructure during the four years from 2025 through 2028 [?].

It follows that ethical and conscious management of these technologies, to mitigate the societal damages they could do while harnessing their potential, is critical. For this very reason, the White House Office of Science and Technology Policy issued a document "AI for Social Good" (*AI4SI*) in 2016 [?], which pioneered this topic among researchers.

